

Infectious disease medicines

An illustrated story + a practical nursing study companion

How to use this guide

Start: Follow the four illustrated story sheets, then use the matching drug cards for the details.

Card key: MOA means mechanism of action: how a drug works. Adverse effects are unwanted effects. A contraindication means do not use in that situation; a caution means evaluate, adjust or monitor.

Scope: Representative, high-yield adult medicines, not an exhaustive formulary. Includes established agents from your earlier HIV table and common current options. Combination, dose, route and duration depend on diagnosis, organ function, pregnancy, susceptibility and the exact product.

Safety: A study reference, not a prescription or a complete interaction checker. A serious allergy to a drug or ingredient generally rules out that product. Check the full label and local protocol before administration.

Every topic in your screenshot

Topics: HIV and AIDS; tuberculosis; transmission-based precautions; *Clostridioides difficile* infection; multidrug-resistant organisms; influenza; viral hepatitis; sepsis and septic shock; Zika virus; West Nile virus; Legionnaires' disease.

Reading map: HIV first. Bacterial illnesses next. Viral illnesses next. Sepsis and precautions last. Acronyms and linked sources close the guide.

Reviewed: 13 September 2026. Sources include federal drug labels, NIH, CDC and specialty guidelines. Older exam examples are identified where relevant.

Same science.
Brighter days. ♥

HIV: Stop the Cell Factory

Human immunodeficiency virus medicines



1 Guard the entrance

Maraviroc blocks CCR5, a cell entry coreceptor.
Fostemsavir blocks attachment.
Ibalizumab blocks post-attachment entry.
CCR5 = C-C chemokine receptor 5

2 Stop the membrane zipper

Enfuvirtide: fusion inhibitor.
Watch: injection-site reactions.

3 Sabotage the copier

Nucleoside/nucleotide reverse transcriptase inhibitors: tenofovir, emtricitabine, lamivudine, abacavir.
Non-nucleoside reverse transcriptase inhibitors: efavirenz, rilpivirine, doravirine.
They stop viral RNA being copied into DNA.

4 Block the inserter

Integrase inhibitors: dolutegravir, bictegravir, raltegravir, cabotegravir.
Check mineral and antacid interactions.

5 Disable the scissors

Protease inhibitors: darunavir, atazanavir.
Block protein cutting and viral maturation.
Check drug interactions.

6 Disrupt the shell

Lenacapavir: capsid inhibitor.
Acts at several stages; treatment requires other active HIV drugs.

KEY SAFETY POINTS

Tenofovir: kidneys and bones

Abacavir: genetic allergy screening

Boosters: interaction check

RNA = ribonucleic acid | DNA = deoxyribonucleic acid



A memory map.
Drug-specific cautions and contraindications are in the companion guide.



HIV | faulty building blocks

At the copying desk, medicines prevent HIV from making a usable DNA copy.

Tenofovir: TDF and TAF

Class / role: Nucleoside/nucleotide reverse transcriptase inhibitor (NRTI). TDF = tenofovir disoproxil fumarate; TAF = tenofovir alafenamide.

MOA: Active tenofovir competes with DNA building blocks and terminates the chain.

Adverse effects: TDF: kidney tubule injury and bone mineral loss. TAF generally has less kidney/bone toxicity, but risk is not zero.

Contraindications / cautions: Caution with kidney disease and nephrotoxic drugs. Renal restrictions differ by formulation and combination. Stopping either can flare hepatitis B.

Nursing checks: Check kidney function, urine protein/glucose when indicated, bone risks and hepatitis B status; verify the exact formulation.

Emtricitabine and lamivudine

Class / role: NRTIs; abbreviated FTC and 3TC.

MOA: Faulty DNA building blocks stop reverse transcription. Also active against hepatitis B.

Adverse effects: Usually tolerated well; headache or nausea. Emtricitabine can cause skin hyperpigmentation. Rare lactic acidosis is a serious class risk.

Contraindications / cautions: Do not combine FTC with 3TC as a duplicate pair. Kidney-based dose rules apply; withdrawal can flare hepatitis B.

Nursing checks: Assess renal function, adherence and viral load. They are components of a regimen, not HIV monotherapy.

Abacavir

Class / role: NRTI; abbreviated ABC.

MOA: Stops elongation of viral DNA.

Adverse effects: Potentially fatal hypersensitivity: fever, rash, gastrointestinal or respiratory symptoms.

Contraindications / cautions: CONTRAINDICATED with HLA-B*5701, previous abacavir hypersensitivity, or moderate/severe hepatic impairment. Never rechallenge after suspected hypersensitivity.

Nursing checks: Confirm genetic test before use; teach multi-system allergy symptoms. HLA = human leukocyte antigen. Consider cardiovascular risk when selecting therapy.

Zidovudine

Class / role: NRTI; abbreviated ZDV or AZT. Older agent with selected uses.

MOA: Stops viral DNA-chain extension.

Adverse effects: Anemia, neutropenia, nausea, myopathy; rare lactic acidosis.

Contraindications / cautions: Caution with low blood counts and other marrow-suppressing medicines.

Nursing checks: Monitor complete blood count and fatigue, pallor or infection signs.

Sources: [Tenofovir](#) | [Abacavir](#) | [HIV](#) | [HIV list](#)

HIV | jam the copier, inserter and scissors

Different tools are blocked at different stages. Similar class names do not mean identical safety rules.

Efavirenz, rilpivirine, doravirine

Class / role: Non-nucleoside reverse transcriptase inhibitors (NNRTIs).

MOA: Bind reverse transcriptase and impair its function rather than acting as DNA building blocks.

Adverse effects: Efavirenz: vivid dreams, dizziness, mood symptoms, rash. Rilpivirine: depressive symptoms and liver effects; rhythm caution. Doravirine: nausea, dizziness.

Contraindications / cautions: Oral rilpivirine: proton pump inhibitors and certain strong enzyme inducers are contraindicated. Many NNRTI interactions are drug-specific.

Nursing checks: Oral rilpivirine requires a meal; acid-reducing drugs need specific review. Efavirenz is an older exam example, not a usual preferred initial option.

Dolutegravir, bictegravir, raltegravir

Class / role: Integrase strand transfer inhibitors (INSTIs).

MOA: Prevent insertion of viral DNA into the cell's DNA.

Adverse effects: Insomnia/headache, weight gain with some regimens; rare hypersensitivity or liver injury. Raltegravir: muscle injury can occur.

Contraindications / cautions: Dolutegravir and bictegravir cannot be used with dofetilide. Rifampin and mineral interactions differ by agent; bictegravir combinations contraindicate rifampin.

Nursing checks: Review antacid, iron and calcium timing. A small creatinine rise with dolutegravir/bictegravir may reflect reduced secretion, not reduced filtration.

Darunavir and atazanavir

Class / role: Protease inhibitors (PIs).

MOA: Block protein-chain cutting, so new HIV cannot mature normally.

Adverse effects: Digestive upset, lipid/glucose changes and liver effects. Atazanavir: indirect bilirubin rise, kidney stones and conduction effects. Darunavir: rash.

Contraindications / cautions: Numerous interacting drugs are contraindicated. Darunavir: caution with sulfonamide allergy. Atazanavir has acid-suppression restrictions.

Nursing checks: Review liver tests, glucose/lipids and all medicines. Take these agents with food as directed; boosting requirements depend on product.

Ritonavir and cobicistat

Class / role: Pharmacokinetic boosters. Ritonavir is also a PI; cobicistat has no direct anti-HIV effect.

MOA: Inhibit drug-metabolizing enzymes, increasing concentrations of a partner drug.

Adverse effects: Digestive effects and major interactions. Cobicistat may raise creatinine without lowering filtration.

Contraindications / cautions: Some sedatives, antiarrhythmics and statins are contraindicated; inhaled fluticasone can cause systemic steroid toxicity.

Nursing checks: Check prescriptions, over-the-counter drugs and supplements before changes. A booster is not an independent complete regimen.

Sources: [Rilpivirine](#) | [Dolutegravir](#) | [HIV](#) | [HIV list](#) | [Cobicistat](#)

HIV | gatekeepers and the capsid

Entry medicines stop different steps at the door. The capsid drug interferes with the protective shell.

Maraviroc

Class / role: CCR5 antagonist. CCR5 = C-C chemokine receptor 5.

MOA: Blocks a human coreceptor used by some HIV strains to enter cells.

Adverse effects: Liver injury with possible allergy features; dizziness and postural hypotension.

Contraindications / cautions: Only for CCR5-tropic HIV. Severe renal impairment plus potent CYP3A inhibitors/inducers is contraindicated.

Nursing checks: Confirm tropism test; check liver tests, blood pressure and interaction-based dosing.

Enfuvirtide

Class / role: Fusion inhibitor; subcutaneous injection.

MOA: Binds gp41 and prevents membrane fusion. gp = glycoprotein.

Adverse effects: Painful injection-site nodules; hypersensitivity; bacterial pneumonia risk.

Contraindications / cautions: Contraindicated after hypersensitivity. Historical teaching example; availability varies.

Nursing checks: Rotate sites and use correct preparation/injection technique; report breathing symptoms.

Fostemsavir

Class / role: Attachment inhibitor; oral prodrug of temsavir.

MOA: Binds viral gp120 to interfere with attachment to CD4.

Adverse effects: Nausea, liver enzyme elevations; heart-rhythm caution in susceptible patients.

Contraindications / cautions: Strong CYP3A inducers such as rifampin are contraindicated.

Nursing checks: Swallow extended-release tablets whole; assess liver tests and interactions.

Sources: [Maraviroc](#) | [Fostemsavir](#) | [Lenacapavir](#) | [HIV list](#)

HIV | gatekeepers and the capsid (continued)

Entry medicines stop different steps at the door. The capsid drug interferes with the protective shell.

Ibalizumab

Class / role: Post-attachment inhibitor; monoclonal antibody given intravenously.

MOA: Binds CD4 and prevents subsequent entry steps.

Adverse effects: Diarrhea, dizziness, nausea/rash; hypersensitivity or infusion reactions.

Contraindications / cautions: Prior hypersensitivity is a contraindication. Used with other active medicines in selected resistant infection.

Nursing checks: Monitor during administration; preserve the scheduled regimen and follow missed-dose instructions.

Lenacapavir

Class / role: Capsid inhibitor.

MOA: Disrupts capsid-dependent processes at multiple stages, including assembly.

Adverse effects: Injection-site reactions/nodules and nausea; prolonged interaction window.

Contraindications / cautions: Strong CYP3A inducers are contraindicated. Treatment and prevention products/regimens are not interchangeable.

Nursing checks: For treatment, use with other active HIV therapy. Follow product-specific injection/loading and missed-dose plans.

Sources: [Maraviroc](#) | [Fostemsavir](#) | [Lenacapavir](#) | [HIV list](#)

HIV and AIDS | the whole treatment plan

Suppress HIV consistently; treat opportunistic infections with medicines directed at their own organisms.

Regimen rules

Core: ART means antiretroviral therapy. Many initial regimens use an integrase inhibitor with two NRTIs; selected validated two-drug regimens also exist. Do not assemble a regimen from the picture.

Long acting: Cabotegravir (integrase inhibitor) plus rilpivirine (NNRTI) is an injectable regimen for selected patients. Injection-site pain, liver/mood effects, interactions and missed-dose resistance risk require planning.

Coinfection: With hepatitis B coinfection, an HIV regimen usually includes two hepatitis B-active agents, commonly tenofovir plus FTC or 3TC. HIV treatment is suppressive, not a cure.

Monitoring: Follow viral load, CD4 count, adherence, kidney/liver function and interactions. CD4 = cluster of differentiation 4, an immune-cell surface protein; CD4 count helps assess immune function.

Trimethoprim-sulfamethoxazole (TMP-SMX)

Class / role: Antifolate combination; prevention/treatment of Pneumocystis pneumonia when indicated.

MOA: Blocks consecutive steps in microbial folate metabolism.

Adverse effects: Rash, severe skin reactions, hyperkalemia, low blood counts and renal effects.

Contraindications / cautions: Avoid after serious sulfonamide/trimethoprim reaction. Pregnancy, renal disease and interacting medicines need clinical review.

Nursing checks: Check complete blood count, potassium and kidney function. Prophylaxis eligibility depends on immune status and viral suppression.

Fluconazole

Class / role: Azole antifungal; common treatment for selected Candida infections.

MOA: Inhibits fungal ergosterol synthesis, impairing cell membranes.

Adverse effects: Digestive upset, liver injury and QT prolongation.

Contraindications / cautions: Some QT-prolonging combinations are contraindicated. Pregnancy and liver disease require review.

Nursing checks: Assess site of infection, renal dose adjustment, liver tests and interactions. Not a substitute for initial combination therapy in severe cryptococcal meningitis.

Severe cryptococcal disease

Treatment: Initial therapy commonly includes liposomal amphotericin B plus flucytosine, followed by fluconazole; regimen varies by setting.

Mechanisms: Amphotericin binds fungal membrane sterols. Flucytosine impairs fungal nucleic-acid synthesis.

Safety: Amphotericin: kidney injury, potassium/magnesium losses and infusion reactions. Flucytosine: marrow/liver toxicity; adjust for kidneys and monitor drug levels where available.

Nursing: Monitor renal function, electrolytes and blood counts. Timing of ART in central nervous system infection requires specialist planning.

Sources: [HIV list](#) | [Opportunistic infections](#) | [Cryptococcosis](#) | [Pneumocystis](#)



1 TUBERCULOSIS: a four-drug team

Different tools. A stronger defense.

Rifampin
STOPS RNA PRODUCTION

Isoniazid
DISRUPTS THE WAXY WALL

Pyrazinamide
DISRUPTS BACTERIAL FUNCTIONS

Ethambutol
BLOCKS WALL BUILDING

MYCOBACTERIUM TUBERCULOSIS

FOUR TOGETHER FOR A STRONGER TOMORROW

Rifampin stops RNA production.
Isoniazid disrupts the waxy wall.
Pyrazinamide disrupts bacterial functions.
Ethambutol blocks wall building.

! Watch the liver, nerves, joints and vision.
Vitamin B₆ protects nerves when indicated.

RNA = ribonucleic acid

2 C. DIFFICILE: treat inside the gut

Clostridioides difficile infection
Target the germs where they live.

Fidaxomicin
BLOCKS BACTERIAL RNA PRODUCTION

Oral Vancomycin
BLOCKS CELL-WALL BUILDING

C. DIFFICILE INFECTION
INFLAMED COLON

ORAL TREATMENT GOES TO THE GUT ✓

DELIVERS TO THE COLON

! Intravenous vancomycin does not replace oral treatment.

! Fulminant illness needs a different treatment plan.

3 RESISTANT GERMS: match the medicine

Use the clues. Choose the right tool.

Look for clues. Match the medicine. Outsmart resistance!

MATCH THE MECHANISM

Vancomycin: cell-wall blocker.

Linezolid: protein-factory blocker.

Daptomycin: membrane disruptor.

CULTURE RESULTS + INFECTION SITE = THE RIGHT CHOICE ✓

DRUG-RESISTANT BACTERIA

GRAM STAIN
CULTURE
SENSITIVITY
SITE MATTERS

RIGHT DRUG
RIGHT PATIENT
BETTER OUTCOMES

! Daptomycin is not for pneumonia.
Culture results and infection site guide the choice.

4 LEGIONNAIRES' DISEASE: reach the hidden bacteria

Get inside to fight back.

Azithromycin
BLOCKS PROTEIN PRODUCTION

Levofloxacin
BLOCKS DNA-COPYING ENZYMES

BACTERIA INSIDE A LUNG IMMUNE CELL (MACROPHAGE)

Legionella pneumophila

! Both can affect heart rhythm.
Levofloxacin also carries tendon and nerve risks.

DNA = deoxyribonucleic acid

DEEPER REACH HEALTHIER BREATHS



Class, mechanism, adverse effects, contraindications and nursing checks: see companion cards.

Same Science. Brighter Futures.

Tuberculosis | the RIPE team

Four workers attack the waxy fortress from different angles. Active tuberculosis needs a combination regimen.

Rifampin (R)

Class / role: Rifamycin antibiotic.

MOA: Inhibits bacterial DNA-dependent RNA polymerase.

Adverse effects: Orange-red body fluids, liver injury, digestive upset; rare platelet problems.

Contraindications / cautions: Strong enzyme induction causes major interactions, including with HIV medicines and hormonal contraception. Specific combinations are contraindicated.

Nursing checks: Teach expected discoloration versus jaundice. Review every medication and contraception; assess liver symptoms.

Isoniazid (I)

Class / role: Antimycobacterial antibiotic.

MOA: Inhibits mycolic-acid synthesis in the bacterial wall.

Adverse effects: Hepatitis and peripheral neuropathy; rare seizures.

Contraindications / cautions: Acute liver disease, prior isoniazid-associated liver injury or serious hypersensitivity are contraindications. Alcohol increases liver risk.

Nursing checks: Give pyridoxine (vitamin B6) when indicated, particularly with neuropathy risks; report numbness and liver symptoms.

Pyrazinamide (P)

Class / role: Antimycobacterial prodrug.

MOA: Converted to pyrazinoic acid; disrupts several bacterial functions. Full mechanism is not completely defined.

Adverse effects: Liver toxicity, increased uric acid and joint pain.

Contraindications / cautions: Contraindicated with severe liver damage or acute gout.

Nursing checks: Check liver status; assess painful joints. Asymptomatic uric-acid elevation is not the same as acute gout.

Ethambutol (E)

Class / role: Antimycobacterial antibiotic.

MOA: Inhibits arabinosyl transferases, disrupting cell-wall construction.

Adverse effects: Optic neuritis: blurred vision and altered red-green discrimination.

Contraindications / cautions: Contraindicated in known optic neuritis unless clinical judgment supports use; caution when vision cannot be assessed and in renal impairment.

Nursing checks: Baseline and ongoing vision/color testing; promptly report changes. Adjust regimen for kidney function.

Remember: RIPE is a mnemonic, not the only regimen. A common drug-susceptible regimen starts with four drugs, then narrows. Shorter regimens exist for eligible patients. Latent TB uses different regimens after active disease is excluded; public-health follow-up and directly observed therapy support completion.

Sources: [TB regimens](#) | [TB safety](#) | [Isoniazid](#) | [Pyrazinamide](#) | [Ethambutol](#)

C. difficile | deliver treatment to the colon

A medicine that reaches the bloodstream is not automatically a medicine that treats the gut lumen.

Fidaxomicin

Class / role: Macrocyclic antibacterial; preferred for many initial/nonfulminant episodes when feasible.

MOA: Inhibits bacterial RNA polymerase, reducing RNA production.

Adverse effects: Nausea, abdominal symptoms; hypersensitivity can occur.

Contraindications / cautions: Contraindicated with hypersensitivity. Evidence does not support replacing the fulminant-disease regimen with fidaxomicin.

Nursing checks: Assess stool frequency, hydration and abdominal findings; monitor for recurrence.

Vancomycin by mouth

Class / role: Glycopeptide antibiotic; alternative for nonfulminant infection and part of fulminant treatment.

MOA: Blocks cell-wall synthesis; oral treatment acts mainly in the gut.

Adverse effects: Nausea; systemic absorption and renal toxicity are possible with severe colitis, especially with renal impairment.

Contraindications / cautions: Serious vancomycin allergy requires review. Intravenous vancomycin alone does not treat luminal C. difficile adequately.

Nursing checks: Verify ROUTE. Oral vancomycin does not treat bloodstream infection. Recurrent disease may need a taper/pulse or other plan.

Metronidazole in fulminant disease

Class / role: Nitroimidazole antibiotic; intravenous adjunct to enteral vancomycin, especially with ileus.

MOA: Reactive metabolites damage microbial DNA.

Adverse effects: Nausea, metallic taste, neuropathy with prolonged exposure and drug interactions.

Contraindications / cautions: Avoid alcohol/propylene glycol during treatment and for the label-specified interval afterward; disulfiram interaction is important.

Nursing checks: Fulminant illness needs urgent multidisciplinary care. Ileus may require rectal vancomycin; monitor abdomen, stool output and systemic deterioration.

Recurrence and prevention

Care: Stop the triggering antibiotic if possible. Correct fluid/electrolyte losses. Recurrent disease may require specialist microbiota-based treatment after antibacterial therapy.

Infection control: Contact precautions, gloves/gown, appropriate spore-killing environmental cleaning and soap-and-water handwashing when indicated. Avoid unprescribed antimotility treatment.

Distinction: Colonization without compatible symptoms is not the same as infection requiring treatment.

Sources: C. difficile | Fulminant C. difficile

Resistant organisms | select for the site

MDRO means multidrug-resistant organism. Resistance is a laboratory and clinical problem, not a single drug indication.

Intravenous vancomycin

Class / role: Glycopeptide; common option for serious susceptible MRSA infection.

MOA: Inhibits cell-wall synthesis. MRSA = methicillin-resistant *Staphylococcus aureus*.

Adverse effects: Kidney injury; infusion reaction with flushing/itching; rare hearing effects.

Contraindications / cautions: True hypersensitivity differs from a rate-related infusion reaction. Caution with other nephrotoxic drugs.

Nursing checks: Monitor kidney function and exposure using the local therapeutic-monitoring protocol. Slow infusion as ordered. It does not cover VRE.

Linezolid

Class / role: Oxazolidinone; selected MRSA infections and VRE. VRE = vancomycin-resistant enterococci.

MOA: Binds the 50S bacterial ribosomal subunit and blocks initiation of protein synthesis.

Adverse effects: Thrombocytopenia/marrow suppression; serotonin syndrome; neuropathy and lactic acidosis with prolonged use.

Contraindications / cautions: Contraindicated with monoamine oxidase inhibitors or within two weeks of use. Serotonergic/adrenergic medicines need an individualized safety plan.

Nursing checks: Monitor blood counts, commonly weekly, and visual/neurologic symptoms. Review antidepressants, opioids and high-tyramine foods.

Daptomycin

Class / role: Cyclic lipopeptide; selected bloodstream/right-sided endocarditis infections; VRE use is specialist-directed.

MOA: Depolarizes bacterial membranes.

Adverse effects: Muscle injury/rhabdomyolysis; eosinophilic pneumonia.

Contraindications / cautions: NOT for pneumonia: lung surfactant inactivates it. Caution with statins and renal impairment.

Nursing checks: Monitor creatine kinase and muscle symptoms, generally at least weekly; assess new respiratory symptoms and renal dosing.

Nursing decision point

Culture: Use the culture, susceptibility pattern, infection site and severity together.

Colonization: A positive screen does not automatically require systemic antibiotics. Decolonization, when indicated, is a separate protocol.

Alternatives: Selected mild skin infections may use other agents such as TMP-SMX or doxycycline if susceptible; this does not make them substitutes for serious bloodstream infection.

Sources: [Linezolid](#) | [Daptomycin](#) | [Resistance](#)

Resistant gram-negative bacteria

The shield differs by organism and resistance enzyme. Specialists match the tool to that shield.

Meropenem

Class / role: Carbapenem beta-lactam antibiotic; common option for serious susceptible ESBL infections.

MOA: Binds bacterial penicillin-binding proteins and blocks cell-wall assembly. ESBL = extended-spectrum beta-lactamase.

Adverse effects: Diarrhea, allergy, seizures/neurotoxicity, C. difficile risk.

Contraindications / cautions: Serious beta-lactam hypersensitivity is a contraindication. Avoid combining with valproate when possible: concentrations can fall sharply.

Nursing checks: Adjust for renal function. It does not cover MRSA and is not reliable for all carbapenem-resistant organisms.

Ceftazidime-avibactam

Class / role: Cephalosporin plus beta-lactamase inhibitor; selected resistant infections.

MOA: Ceftazidime blocks the wall; avibactam protects it from certain resistance enzymes.

Adverse effects: Digestive effects, allergy and neurotoxicity with excess exposure.

Contraindications / cautions: Serious component/cephalosporin allergy is a contraindication. Alone it does not cover metallo-beta-lactamase producers.

Nursing checks: Renal dosing and resistance testing are essential. Combination with aztreonam may be selected for particular resistance mechanisms.

Cefiderocol

Class / role: Siderophore cephalosporin; selected difficult-to-treat gram-negative infections.

MOA: Uses bacterial iron-uptake pathways to enter, then inhibits cell-wall synthesis.

Adverse effects: Allergy, diarrhea and liver-test changes; mortality warning from a trial of critically ill patients with resistant infections.

Contraindications / cautions: Serious beta-lactam hypersensitivity is a contraindication; use requires careful organism/site-specific assessment.

Nursing checks: Monitor clinical response, renal function and susceptibility. Not a general replacement for standard antibiotics.

Decode the resistance names

CRE: Carbapenem-resistant Enterobacterales: a group with different resistance mechanisms, not one uniform treatment.

DTR: Difficult-to-treat resistance; used for organisms such as Pseudomonas aeruginosa.

Rule: There is no universal MDRO antibiotic. Other newer combinations may be preferable for a specific resistance enzyme; this page gives representative mechanisms.

Sources: [Resistance](#) | [Cefiderocol](#) | [Meropenem](#) | [Ceftazidime/avibactam](#)

Legionnaires' disease | reach inside the cell

Legionella can live inside immune cells. The selected antibiotic needs useful intracellular activity.

Azithromycin

Class / role: Macrolide antibiotic; common Legionella-directed option.

MOA: Binds the 50S ribosomal subunit and blocks bacterial protein production.

Adverse effects: Nausea/diarrhea, liver injury, QT prolongation; hearing effects are uncommon.

Contraindications / cautions: Contraindicated after azithromycin-associated cholestatic jaundice/hepatic dysfunction or serious macrolide allergy. Caution with arrhythmia risks.

Nursing checks: Review electrocardiogram risks, potassium/magnesium, interacting drugs and liver symptoms.

Levofloxacin

Class / role: Respiratory fluoroquinolone; another common option.

MOA: Inhibits bacterial DNA gyrase and topoisomerase IV, disrupting DNA replication.

Adverse effects: Tendon injury/rupture, peripheral neuropathy, central nervous system effects, glucose disturbances, QT prolongation and C. difficile.

Contraindications / cautions: Contraindicated with quinolone hypersensitivity. Avoid in myasthenia gravis; assess aortic aneurysm/dissection risk and other major warnings.

Nursing checks: Adjust for renal function; separate oral doses from iron, calcium supplements and antacids per label. Report tendon pain or new numbness promptly.

Clinical context

Choice: Azithromycin or a respiratory fluoroquinolone is commonly used; severity, organ function and interactions guide selection.

Testing: Obtain appropriate respiratory testing and urinary antigen testing without delaying treatment.

Distinction: Pontiac fever is a different, self-limited Legionella syndrome and generally does not need antibiotics.

Sources: [Legionella](#) | [Levofloxacin](#) | [Azithromycin](#)

DIFFERENT
VIRUSES
DIFFERENT JOBS
BRIGHTER
TOMORROWS

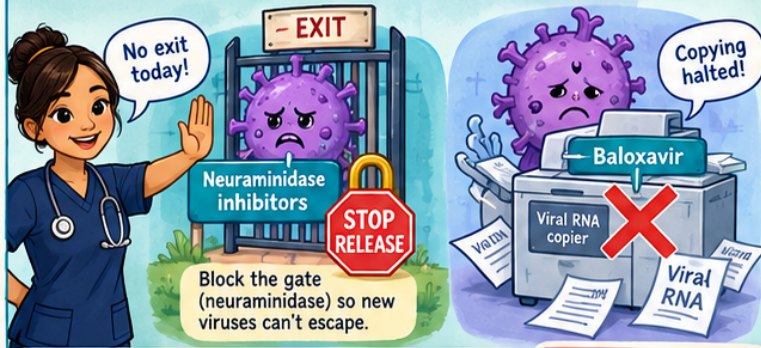
Viruses: Different Jobs, Different Medicines

Same goal: understand the virus, choose the right support, make a difference.

Nurses
Connect
Science
to Better Days

INFLUENZA: block the escape or the copier

Stop new viruses from leaving or copying their genetic material.



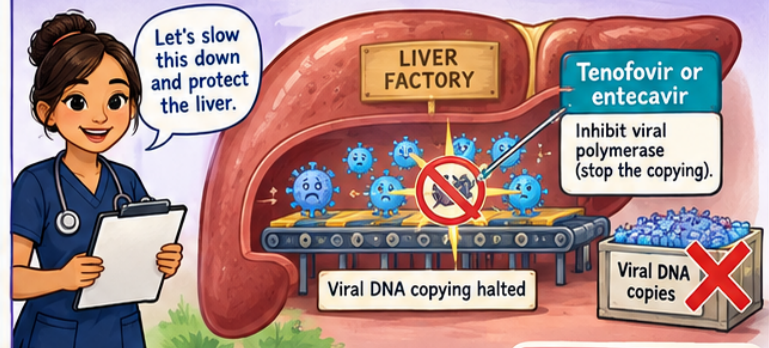
Oseltamivir, zanamivir, peramivir: neuraminidase inhibitors; reduce release of new viruses.

Baloxavir: blocks viral RNA transcription.

SAFETY
Zanamivir:
bronchospasm risk.
Oseltamivir:
nausea and vomiting.

HEPATITIS B: suppress the copying

Stop the virus from making more copies. This controls the infection but does not cure it.



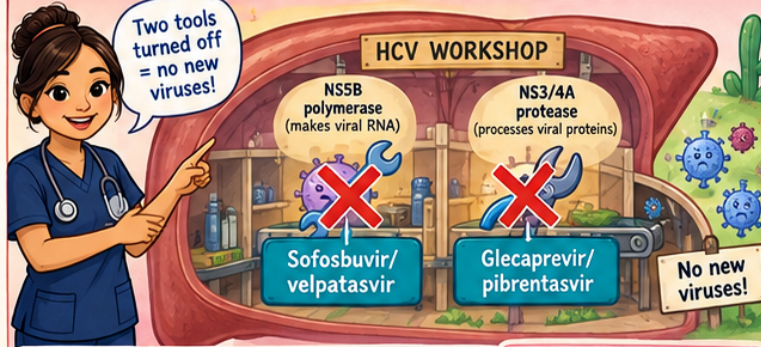
Tenofovir or entecavir: inhibit viral polymerase.

Often long-term treatment; do not stop without a plan.

SAFETY
Check kidneys.
Stopping can cause hepatitis flares.

HEPATITIS C: block two viral tools

Stop the virus from copying and building new viruses.



Sofosbuvir/velpatasvir
Glecaprevir/pibrentasvir

Combination antivirals can cure hepatitis C.

SAFETY
Screen for hepatitis B reactivation.
Glecaprevir/pibrentasvir: avoid with decompensated liver disease.

ZIKA + WEST NILE: support the patient

No specific antiviral treatment. Help the body recover and monitor for severe illness.



SAFETY
Suspected Zika: avoid aspirin and anti-inflammatory pain medicines until dengue is excluded.



Hepatitis A: supportive care.

Vaccines prevent some infections; they do not treat established illness.



Mini glossary:

RNA = ribonucleic acid

Mechanisms and safety details continue in the companion drug cards.

KNOW
THE VIRUS.
SUPPORT
THE PATIENT.
THAT'S NURSING.

Influenza | shut the exit or stop transcription

Treat promptly when indicated. Hospitalized or high-risk patients may benefit even when more than 48 hours have passed.

Oseltamivir

Class / role: Oral neuraminidase inhibitor.

MOA: Reduces release of new influenza A and B viruses.

Adverse effects: Nausea/vomiting, headache; rare serious skin or neuropsychiatric events.

Contraindications / cautions: Serious hypersensitivity is a contraindication; renal adjustment may be needed.

Nursing checks: Give with food if nauseated. Commonly preferred for hospitalized patients and during pregnancy.

Zanamivir

Class / role: Inhaled neuraminidase inhibitor.

MOA: Reduces release of new influenza viruses.

Adverse effects: Bronchospasm, cough; rare allergy.

Contraindications / cautions: Contraindicated with allergy to milk proteins in the inhalation product. Not recommended in asthma or chronic obstructive pulmonary disease.

Nursing checks: Check ability to use the device and airway history. Do not nebulize the dry-powder product.

Peramivir

Class / role: Intravenous neuraminidase inhibitor.

MOA: Inhibits release of new influenza viruses.

Adverse effects: Diarrhea; rare serious skin/allergic and neuropsychiatric reactions.

Contraindications / cautions: Serious hypersensitivity is a contraindication; renal adjustment is important.

Nursing checks: Used for selected patients; not routinely the preferred hospitalized-patient regimen.

Baloxavir

Class / role: Cap-dependent endonuclease inhibitor; oral antiviral.

MOA: Blocks the cap-snatching step needed for viral RNA transcription.

Adverse effects: Usually well tolerated; hypersensitivity is possible.

Contraindications / cautions: Not recommended during pregnancy/breastfeeding or for hospitalized, complicated illness; avoid coadministration with polyvalent minerals.

Nursing checks: Avoid taking with dairy/calcium-fortified drinks, antacids or mineral supplements. Age and clinical eligibility matter.

Remember: Amantadine and rimantadine are not recommended for current seasonal influenza because of widespread resistance. Vaccination prevents disease; it is not treatment for an established episode.

Sources: [Flu](#) | [Flu choices](#) | [Zanamivir](#)

Viral hepatitis | A, B and the liver

Hepatitis means liver inflammation. The letter identifies a different virus, not a stage of the same disease.

Tenofovir for chronic hepatitis B

Class / role: Nucleotide analogue polymerase inhibitor; TDF or TAF in appropriate patients.

MOA: Suppresses hepatitis B viral DNA synthesis.

Adverse effects: TDF: kidney and bone toxicity; TAF generally less of these effects. Severe hepatitis flares can follow withdrawal.

Contraindications / cautions: Kidney/liver restrictions depend on formulation. HIV coinfection requires a fully suppressive HIV regimen, not hepatitis B monotherapy.

Nursing checks: Monitor renal/liver tests and viral response. Do not stop without a monitoring plan.

Entecavir

Class / role: Guanosine nucleoside analogue; hepatitis B polymerase inhibitor.

MOA: Inhibits several steps of viral DNA synthesis.

Adverse effects: Headache, fatigue, dizziness; rare lactic acidosis, especially with advanced disease. Withdrawal flares.

Contraindications / cautions: No specific label contraindications, but untreated HIV coinfection risks HIV resistance. Renal impairment needs adjustment.

Nursing checks: Take on an empty stomach according to label; test for HIV before monotherapy; follow liver function and viral DNA.

Peginterferon alfa-2a

Class / role: Immune-modulating interferon; selected hepatitis B patients.

MOA: Promotes antiviral immune responses.

Adverse effects: Flu-like symptoms, depression/suicidality, marrow suppression, thyroid/autoimmune problems.

Contraindications / cautions: Contraindicated with autoimmune hepatitis or hepatic decompensation; major psychiatric and autoimmune cautions.

Nursing checks: Monitor mood, blood counts, liver and thyroid tests. Not appropriate for every patient.

A, D and E: different plans

A: Hepatitis A usually receives supportive care. Vaccine and, for selected exposures, immune globulin are preventive measures.

D: Hepatitis D requires hepatitis B infection; specialist-directed treatment and hepatitis B prevention are important.

E: Hepatitis E is often managed supportively. Pregnancy and chronic infection in immunosuppressed patients require specialist assessment; do not assume the hepatitis B or C drugs apply.

Sources: [Hepatitis B](#) | [Entecavir](#) | [Tenofovir](#) | [Peginterferon](#)

Hepatitis C | stop two viral tools

Direct-acting antiviral combinations can cure hepatitis C. A cure does not prevent a new infection.

Sofosbuvir / velpatasvir

Class / role: NS5B polymerase inhibitor + NS5A inhibitor. NS means nonstructural viral protein.

MOA: Sofosbuvir blocks RNA synthesis; velpatasvir disrupts replication/assembly functions.

Adverse effects: Headache, fatigue; hepatitis B reactivation is a major warning.

Contraindications / cautions: Amiodarone coadministration is not recommended because of severe bradycardia risk. Acid suppressants and enzyme inducers need review.

Nursing checks: Screen for hepatitis B, assess liver status and interactions, confirm adherence and document sustained viral response after therapy.

Glecaprevir / pibrentasvir

Class / role: NS3/4A protease inhibitor + NS5A inhibitor.

MOA: Blocks viral protein processing and replication/assembly.

Adverse effects: Headache, fatigue, nausea; hepatitis B reactivation and liver decompensation warnings.

Contraindications / cautions: CONTRAINDICATED in moderate/severe hepatic impairment or any prior hepatic decompensation, and with rifampin or atazanavir.

Nursing checks: Take with food. Assess liver history, hepatitis B status and the complete medication list.

Ribavirin: selected specialist regimens

Class / role: Nucleoside analogue antiviral; not routine monotherapy for hepatitis C.

MOA: Has several antiviral effects, including interference with viral RNA processes.

Adverse effects: Hemolytic anemia; marked embryo-fetal toxicity.

Contraindications / cautions: CONTRAINDICATED in pregnancy and in men whose partners are pregnant; also important cardiac, hemoglobinopathy and renal restrictions.

Nursing checks: Pregnancy testing and effective contraception during and for the label-specified months after treatment; check blood counts and renal function.

Terms in the illustration

Polymerase: An enzyme that builds nucleic-acid chains.

NS5B: A hepatitis C RNA polymerase.

NS5A: A hepatitis C protein involved in replication and assembly.

NS3 4A: A hepatitis C protease complex. These are viral-protein names, not nursing assessment scales.

Sources: [Hepatitis C](#) | [Sofosbuvir/velpatasvir](#) | [Glecaprevir/pibrentasvir](#) | [Ribavirin](#)

Zika and West Nile | support, observe, protect

No approved specific antiviral cures these infections. Symptom medicines support the patient rather than eliminate the virus.

Acetaminophen: symptom relief

Class / role: Analgesic and antipyretic; pain/fever medicine.

MOA: Acts mainly through central pathways to reduce pain and fever; exact mechanism is not fully defined.

Adverse effects: Liver injury with excessive total exposure; rare serious skin reactions.

Contraindications / cautions: Severe active liver disease and product-specific restrictions require review. Avoid duplicate acetaminophen-containing products.

Nursing checks: Check the total from all products and liver/alcohol history. This is symptom relief, not antiviral treatment.

Zika virus

Treatment: Rest, hydration and symptom relief. No specific antiviral or preventive vaccine is available in current CDC guidance.

Avoid: Avoid aspirin and nonsteroidal anti-inflammatory drugs until dengue is excluded because bleeding risk changes management.

Nursing: Assess pregnancy/exposure history; coordinate follow-up for pregnancy. Teach mosquito and sexual-transmission prevention.

West Nile virus

Treatment: Supportive care. Severe neurologic disease may require hospital care, airway support, fluids and treatment of seizures or other complications.

Caution: Antibiotics do not treat the virus; use them only for another suspected/confirmed bacterial infection. No proven routine antiviral regimen.

Nursing: Watch mental status, weakness, swallowing and breathing. Prevention centers on avoiding mosquito bites.

Sources: [Zika](#) | [West Nile](#)



Sepsis Rescue + Stop the Spread

TREAT THE PATIENT. PROTECT OTHERS.



1 Find and treat the infection

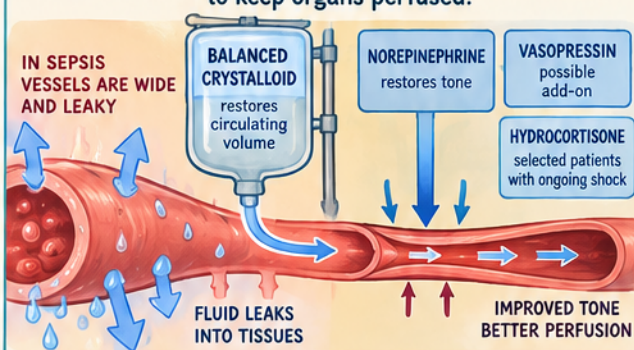
Evaluate for the source and collect cultures while giving antibiotics concurrently.



- Antibiotics are chosen for the source, likely germs and resistance risk.
- Examples: cefepime, piperacillin/tazobactam, meropenem; add vancomycin when resistant staphylococcal coverage is needed.
- Obtain cultures promptly without delaying urgent antibiotics.

2 Support the circulation

Restore volume and maintain blood pressure to keep organs perfused.



- Balanced crystalloid: restores circulating volume.
- Norepinephrine: first-line vasopressor.
- Vasopressin: possible add-on.
- Hydrocortisone: selected patients with ongoing shock.

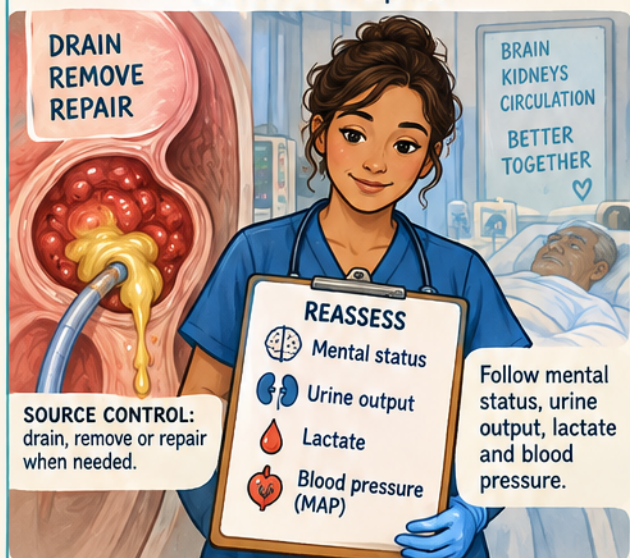


SAFETY

- Reassess fluids. Pressors may start during fluids in unstable shock.
- Watch for fluid overload, arrhythmias and tissue ischemia.

3 Remove the source; reassess

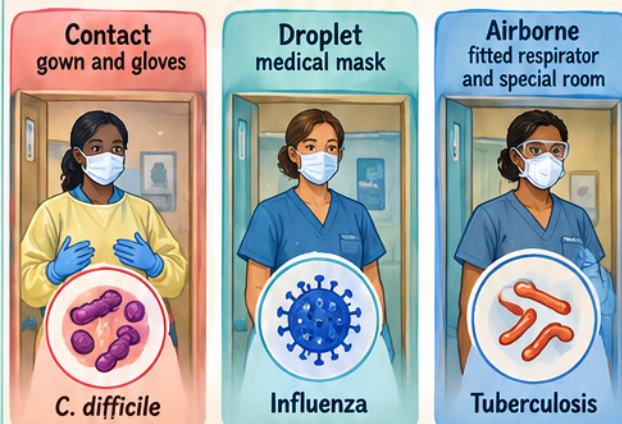
Control the infection at its source and follow for response.



MAP = mean arterial pressure

4 Precautions are a separate layer of protection

Infection control helps stop the spread of germs in health care settings.



Tuberculosis: airborne. Influenza: droplet. C. difficile: contact.

Standard precautions apply to everyone.

TB = tuberculosis | C. difficile = Clostridioides difficile



Precautions are not a drug class.
Antibiotics do not replace infection control.



SAFER
CARE
BRIGHTER
TOMORROWS

Sepsis | restore circulation

The team treats infection and supports blood flow at the same time. Medicines are titrated to response.

Balanced crystalloids, such as lactated Ringer's

Class / role: Intravenous fluid, not an antimicrobial.

MOA: Expands extracellular and circulating volume.

Adverse effects: Fluid overload, edema and electrolyte/acid-base changes.

Contraindications / cautions: Individualize in heart/kidney dysfunction; no automatic unlimited bolus. Compatibility depends on the solution and co-infused drug.

Nursing checks: Frequently reassess perfusion, lungs, urine output and fluid responsiveness. The adult guideline suggests at least 30 mL/kg initially within 3 hours for septic shock/hypoperfusion, individualized.

Norepinephrine

Class / role: First-line vasopressor; adrenergic agonist.

MOA: Mainly alpha-1 vessel constriction, with some beta-1 cardiac support.

Adverse effects: Peripheral/organ ischemia, arrhythmias and injury if it extravasates.

Contraindications / cautions: Correct volume depletion as appropriate, but unstable shock may need pressors during fluids. Major interactions with certain antidepressants and anesthetics.

Nursing checks: Continuous pressure/rhythm assessment, frequent site checks and perfusion checks. Peripheral initiation may avoid delay under protocol. Extravasation needs immediate protocol-based treatment.

Vasopressin

Class / role: Non-catecholamine vasopressor; adjunct.

MOA: Activates vascular V1 receptors to tighten vessels.

Adverse effects: Digital/mesenteric ischemia, reduced cardiac output and sodium disturbances.

Contraindications / cautions: Hypersensitivity is a contraindication; caution in ischemic disease.

Nursing checks: Monitor perfusion, urine output and sodium; usually used as an adjunct rather than replacing initial norepinephrine.

Epinephrine

Class / role: Adrenergic vasopressor/inotrope; selected add-on.

MOA: Stimulates alpha and beta receptors to support pressure and cardiac output.

Adverse effects: Tachyarrhythmias, hyperglycemia, ischemia and lactate rise.

Contraindications / cautions: Drug interactions and cardiac risk require monitoring; emergency use is indication-dependent.

Nursing checks: Interpret lactate in context: epinephrine itself can raise it. Track rhythm, glucose, pressure and organ perfusion.

Sources: [Sepsis](#) | [Sepsis quick guide](#)

Sepsis | attack the source and reassess

Antibiotic selection depends on likely source, resistance risk, allergy and organ function. These are examples, not a universal combination.

Piperacillin / tazobactam

Class / role: Antipseudomonal penicillin + beta-lactamase inhibitor.

MOA: Blocks wall synthesis; tazobactam protects against some bacterial enzymes.

Adverse effects: Allergy, diarrhea/C. difficile, electrolyte effects and renal concerns.

Contraindications / cautions: Serious beta-lactam allergy is a contraindication. Does not cover MRSA; not the preferred choice for every resistant organism.

Nursing checks: Renal dosing, compatibility, sodium burden and cultures. Narrow treatment as information returns.

Cefepime

Class / role: Fourth-generation cephalosporin.

MOA: Blocks bacterial cell-wall synthesis.

Adverse effects: Encephalopathy, myoclonus or seizures, especially with renal accumulation; allergy and diarrhea.

Contraindications / cautions: Immediate serious beta-lactam hypersensitivity is a contraindication. It lacks reliable anaerobic and MRSA coverage.

Nursing checks: Renal adjustment and mental-status monitoring are critical. Add other coverage only when the source/risk indicates it.

Hydrocortisone

Class / role: Corticosteroid; adjunct in selected septic shock.

MOA: Modulates inflammation and vascular responsiveness.

Adverse effects: Hyperglycemia, fluid/sodium effects, infection risk, weakness and gastrointestinal effects.

Contraindications / cautions: Systemic fungal infection is a usual label contraindication except selected circumstances; weigh risks in shock. Not an antibiotic.

Nursing checks: Monitor glucose, sodium, potassium, infection and response; use the prescribed course/taper plan.

What makes treatment complete

Other agents: Meropenem may be selected for resistant gram-negative risk; add intravenous vancomycin for appropriate MRSA risk. See their cards for safety.

Source control: Drain an abscess, remove an infected device or correct the source when indicated.

Reassess: Obtain cultures promptly without delaying urgent therapy; use results to narrow/stop antibiotics. Track mental status, urine output, perfusion and lactate trends.

Pressure: MAP = mean arterial pressure. An initial adult target is commonly about 65 mmHg; the 2026 guideline suggests 60-65 in adults 65 or older. Individualize clinically.

Sources: Sepsis | Sepsis quick guide | Cefepime | Resistance | Piperacillin/tazobactam | Hydrocortisone

Transmission-based precautions

The nurse puts a barrier between the germ and the next patient. These are infection-control measures, not medicines.

Contact

Examples: C. difficile and selected resistant-organism situations.

Actions: Gown/gloves, dedicated equipment where possible, appropriate room placement and environmental cleaning. For C. difficile, use a sporicidal product and soap-and-water handwashing as indicated.

Drug link: Treat the infection when indicated, but do not assume an antibiotic immediately ends infectiousness.

Droplet

Example: Seasonal influenza.

Actions: Medical mask on room entry and other protection based on exposure; patient source control during transport. Procedures may require additional respiratory protection.

Drug link: Antiviral treatment does not replace droplet precautions.

Airborne

Example: Suspected or confirmed infectious pulmonary/laryngeal tuberculosis.

Actions: Airborne infection isolation room and fit-tested respirator, such as N95 or higher-level protection. N95 is a respirator classification, not a medication.

Drug link: Treatment plus clinical/microbiologic and infection-control criteria guide discontinuation of isolation.

Standard precautions apply to everyone

Practice: Hand hygiene, injection/sharps safety, respiratory hygiene and exposure-based protective equipment.

Context: HIV, hepatitis B/C, Zika, West Nile and Legionnaires' disease generally do not require added isolation solely for that diagnosis; symptoms or coinfections can change precautions.

Prevention: Vaccines and post-exposure prophylaxis have separate indications. Hepatitis A/B and influenza vaccines prevent their respective infections; none is a cure for established disease.

Sources: [Precautions](#)

Acronym decoder | drugs and mechanisms

The same letters appear on several cards. Read the full name once, then connect it to the job.

Infections and treatment

HIV: Human immunodeficiency virus.

AIDS: Acquired immunodeficiency syndrome, the most advanced stage of HIV infection.

ART: Antiretroviral therapy.

TB: Tuberculosis.

HBV HCV: Hepatitis B virus / hepatitis C virus.

MOA: Mechanism of action: how a medicine produces its effect.

OI: Opportunistic infection: infection that takes advantage of impaired defenses.

HIV drug classes

NRTI: Nucleoside/nucleotide reverse transcriptase inhibitor.

NNRTI: Non-nucleoside reverse transcriptase inhibitor.

INSTI: Integrase strand transfer inhibitor.

PI: Protease inhibitor.

TDF TAF: Tenofovir disoproxil fumarate / tenofovir alafenamide.

FTC 3TC: Emtricitabine / lamivudine.

ABC ZDV AZT: Abacavir / zidovudine / another abbreviation for zidovudine.

Targets and tests

DNA RNA: Deoxyribonucleic acid / ribonucleic acid: genetic materials.

CD4: Cluster of differentiation 4; immune-cell receptor and marker.

CCR5: C-C chemokine receptor 5; HIV entry coreceptor.

gp: Glycoprotein; gp120 and gp41 are HIV envelope proteins.

HLA B5701: Human leukocyte antigen B*5701 genetic variant; predicts increased abacavir hypersensitivity risk.

CYP3A: Cytochrome P450 3A enzymes: a major drug-metabolism pathway. Inducers can lower levels of certain medicines; inhibitors can raise them.

Acronym decoder | bedside safety

A caution needs a plan; a contraindication rules out use in the stated situation.

Resistant organisms

MDRO: Multidrug-resistant organism.

MRSA: Methicillin-resistant *Staphylococcus aureus*.

VRE: Vancomycin-resistant enterococci.

ESBL: Extended-spectrum beta-lactamase, a resistance enzyme.

CRE: Carbapenem-resistant Enterobacterales.

DTR: Difficult-to-treat resistance.

Monitoring and administration

IV: Intravenous: into a vein.

CBC: Complete blood count.

CK: Creatine kinase, a muscle-injury marker.

QT: An electrocardiogram interval covering ventricular electrical activation and recovery. Prolongation can raise arrhythmia risk.

MAP: Mean arterial pressure.

mmHg: Millimeters of mercury, a pressure unit.

mL kg: Milliliters per kilogram.

TMP SMX: Trimethoprim-sulfamethoxazole.

NSAID: Nonsteroidal anti-inflammatory drug, such as ibuprofen.

Plain-language safety terms

Nephrotoxicity: Kidney injury.

Hepatotoxicity: Liver injury.

Neuropathy: Nerve damage, often with tingling, numbness or pain.

Myelosuppression: Reduced blood-cell production in bone marrow.

Ischemia: Inadequate blood flow to tissue.

Extravasation: Infused drug escapes the vein into surrounding tissue.

Decompensated liver disease: Liver disease with complications such as fluid accumulation, confusion or variceal bleeding.

Prophylaxis: Prevention. Empiric treatment starts before the exact organism is known; targeted treatment uses diagnostic results.

Sources and further checking

Click a source title to open its guideline or prescribing information. Clinical recommendations change; use the current product label for exact contraindications, interactions and dosing.

HIV

clinicalinfo.hiv.gov

HIV list

hivinfo.nih.gov

Abacavir

dailymed.nlm.nih.gov

Tenofovir

dailymed.nlm.nih.gov

Dolutegravir

dailymed.nlm.nih.gov

Rilpivirine

www.dailymed.nlm.nih.gov

Maraviroc

dailymed.nlm.nih.gov

Fostemsavir

dailymed.nlm.nih.gov

Lenacapavir

dailymed.nlm.nih.gov

TB regimens

www.cdc.gov

TB safety

www.cdc.gov

Isoniazid

dailymed.nlm.nih.gov

C. difficile

www.idsociety.org

Fulminant C. difficile

www.idsociety.org

Resistance

www.idsociety.org

Linezolid

dailymed.nlm.nih.gov

Sources and further checking

Click a source title to open its guideline or prescribing information. Clinical recommendations change; use the current product label for exact contraindications, interactions and dosing.

Daptomycin

dailymed.nlm.nih.gov

Legionella

www.cdc.gov

Levofloxacin

dailymed.nlm.nih.gov

Flu

www.cdc.gov

Flu choices

www.cdc.gov

Hepatitis B

www.cdc.gov

Entecavir

dailymed.nlm.nih.gov

Hepatitis C

www.hcvguidelines.org

Glecaprevir/pibrentasvir

dailymed.nlm.nih.gov

Sofosbuvir/velpatasvir

www.dailymed.nlm.nih.gov

Zika

cdc.gov

West Nile

www.cdc.gov

Sepsis

www.sccm.org

Sepsis quick guide

sccm.org

Cefepime

dailymed.nlm.nih.gov

Precautions

www.cdc.gov

Sources and further checking

Click a source title to open its guideline or prescribing information. Clinical recommendations change; use the current product label for exact contraindications, interactions and dosing.

Opportunistic infections

clinicalinfo.hiv.gov

Cryptococcosis

www.cdc.gov

Pyrazinamide

dailymed.nlm.nih.gov

Ethambutol

dailymed.nlm.nih.gov

Azithromycin

dailymed.nlm.nih.gov

Zanamivir

dailymed.nlm.nih.gov

Cobicistat

dailymed.nlm.nih.gov

Cefiderocol

www.dailymed.nlm.nih.gov

Piperacillin/tazobactam

dailymed.nlm.nih.gov

Hydrocortisone

dailymed.nlm.nih.gov

Ribavirin

dailymed.nlm.nih.gov

Meropenem

www.dailymed.nlm.nih.gov

Ceftazidime/avibactam

dailymed.nlm.nih.gov

Pneumocystis

clinicalinfo.hiv.gov

Peginterferon

dailymed.nlm.nih.gov